

Chemlon® 133 GH

Teknor Apex Company - Polyamide 66

Thursday, May 19, 2022

General Information

Product Description

Chemlon® 133 GH is a 33% glass fiber reinforced, heat stabilized polyamide 66 (PA 66) designed for injection molding. This high tensile strength material has a wide processing window, provides a good surface appearance, and is available globally.

General

Material Status	• Commercial: Active		
Availability	• Africa & Middle East • Asia Pacific	• Europe • Latin America	• North America
Additive	• Heat Stabilizer		
Features	• Good Processability • Good Surface Finish	• Good Thermal Stability • High Tensile Strength	
RoHS Compliance	• Contact Manufacturer		
Automotive Specifications	• 3M 11-0003-5762-1 ¹ • CHRYSLER MS-DB-41 CPN1900 Color: BK001 Black ¹ • CHRYSLER MS-DB-41 CPN2224 Color: Non-matched Color ¹ • FORD ESB-M4D133-A ¹ • FORD ESB-M4D89-A ¹ • FORD WSK-M4D663-A ¹ • GM GMP.PA66.013 Color: Natural ¹ • GM GMW3038P-PA66-GF35H ¹		
Forms	• Pellets		
Processing Method	• Injection Molding		

ASTM & ISO Properties ²

Physical	Dry	Conditioned	Unit	Test Method
Density	1.38	--	g/cm ³	ISO 1183
Molding Shrinkage				ISO 294-4
Across Flow	0.30	--	%	
Flow	0.10	--	%	
Mechanical	Dry	Conditioned	Unit	Test Method
Tensile Modulus	13600	8670	MPa	ISO 527-2
Tensile Stress	189	126	MPa	ISO 527-2
Tensile Strain (Break)	2.4	4.8	%	ISO 527-2
Flexural Modulus	9000	6840	MPa	ISO 178
Flexural Stress	275	169	MPa	ISO 178
Impact	Dry	Conditioned	Unit	Test Method
Charpy Notched Impact Strength	6.6	8.6	kJ/m ²	ISO 179
Charpy Unnotched Impact Strength	53	70	kJ/m ²	ISO 179
Notched Izod Impact Strength 23°C	7.5	9.2	kJ/m ²	ISO 180
Thermal	Dry	Conditioned	Unit	Test Method
Deflection Temperature Under Load 0.45 MPa, Unannealed	255	--	°C	ISO 75-2/B
Deflection Temperature Under Load 1.8 MPa, Unannealed	249	--	°C	ISO 75-2/A
Melting Temperature	259	--	°C	
CLTE - Flow	5.0E-5	--	cm/cm/°C	ASTM D696

Revision Date: 7/31/2020

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Thermal	Dry	Conditioned	Unit	Test Method
RTI Elec				UL 746B
0.8 mm	130	--	°C	
1.5 mm	130	--	°C	
3.0 mm	130	--	°C	
RTI Imp				UL 746B
0.8 mm	95.0	--	°C	
1.5 mm	110	--	°C	
3.0 mm	110	--	°C	
RTI Str				UL 746B
0.8 mm	130	--	°C	
1.5 mm	130	--	°C	
3.0 mm	130	--	°C	
Electrical	Dry	Conditioned	Unit	Test Method
Volume Resistivity	1.0E+14	--	ohms·cm	ASTM D257
Dielectric Strength (3.00 mm)	16	--	kV/mm	ASTM D149
Comparative Tracking Index (CTI)	600	--	V	UL 746A
Flammability	Dry	Conditioned	Unit	Test Method
Flame Rating (0.8 mm)	HB	--		UL 94
Oxygen Index	25	--	%	ISO 4589-2

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Processing Information		
Injection	Dry	Unit
Drying Temperature	80	°C
Suggested Max Moisture	0.13	%
Processing (Melt) Temp	265 to 293	°C
Mold Temperature	77 to 88	°C

Injection Notes
Maximum peak injection pressure should not exceed 80% of the machine's maximum pressure capability. Start with a holding pressure that is half the peak injection pressure. Perform a rheology curve in order to determine appropriate injection rate.

Notes

¹ Automotive site approvals apply for US manufactured compound only

² Typical properties: these are not to be construed as specifications.

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Teknor Apex Company Corporate Headquarters

*In U.S. for Vinyls, TPEs, Colorants,
Engineered Thermoplastics (Chem Polymer)*
505 Central Avenue
Pawtucket, Rhode Island 02861 U.S.

Phone: 401-725-8000
Fax: 401-725-8095
Toll Free (U.S. only) 800-556-3864

www.teknorapex.com
info@teknorapex.com

Teknor Apex B.V.

Mijnweg 1,
6167 AC Geleen, Netherlands

Phone: +31 46 7020 950
Fax: +31 46 7020 990

www.teknorapex.com
tpe@teknorapex.com

Teknor Apex (Suzhou) Advanced Polymer Compounds Co. Pte. Ltd.

No. 78 Ping Sheng Road
Suzhou Industrial Park
Jiangsu, China 215126

Phone: (86) 512-6287-1550
Fax: (86) 512-6288-8371

www.teknorapex.com
infotaap@teknorapex.com

Teknor Apex Asia Pacific PTE. LTD.

41 Shipyard Road
Singapore 628134

Phone: (65) 6265-2544
Fax: (65) 6265-1821

www.teknorapex.com
infotaap@teknorapex.com

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